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Chapter #4

Learning from Data

Learning from data & Biological inspiration:

Biological learning:

- Biological system, including humans, learn to handle the unpredictable environment through data-driven experiences.

Ex:

- Babies learn to walk without knowing machines.

Predictive learning process:

⇒ Two main phases of predictive learning phase:

1. Learning/Estimating dependencies:

- Use a set of samples to learn unknown system dependencies.

a) Predicting new outputs:

- Use the estimated dependencies to predict outputs for new input values.

⇒ These two phases corresponds to:

Induction:

- Progress from specific cases to a general model.

Deduction:

- Apply the general model specific cases.

Global Vs Local function estimation:

Global function estimation:

- A single model applied to all possible input values.
- Can be excessive for some practical problems.

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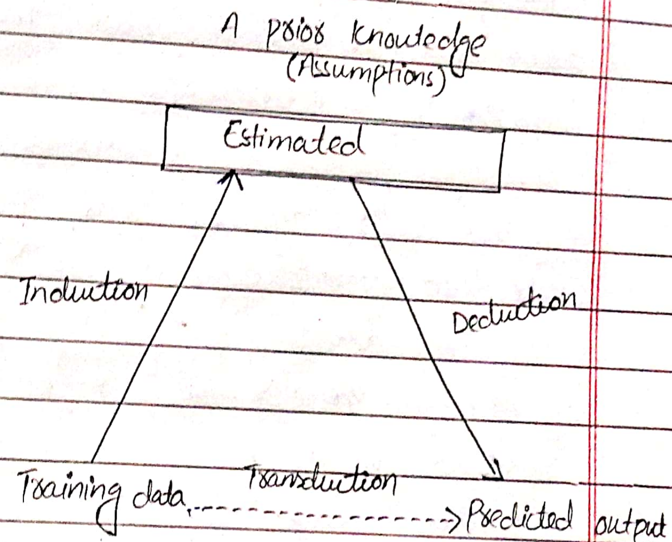
Transductive inference:

- Focuses on estimating outputs directly for specific points of interest from training data.

Types of inference

⇒ Three types of inference:

- Induction
- Deduction
- Transduction



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Q Analyze the General model of inductive learning process?

- Inductive learning involves deriving general principles from specific examples or data points.
- In observational environment, means learning patterns, relationships and dependencies from observed data without explicit instruction.

Key components

- Data Collection
- Feature extraction
- model building
- Training Phase
- model evaluation
- feedback loop

—(elaboration (yy))

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Inductive learning process in observational environment

- Observation
- Pattern recognition
- Generalization
- Application

—(elaboration (yy))

Advantages:

- Data-driven
- flexibility
- Continuous improvement.

Disadvantages:

- Data quality
- Complexity
- Overfitting

Example:

- Weather prediction
- Healthcare
- Finance

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Learning Machine

Machine learning overview:

- Combination of AI and statistics.
- Use various algorithms to solve problems.
- Algorithms vary in goals, training data, learning strategies, and data representation.

Learning process:

- All algorithms search an n-dimensional space of data to find a generalization.
- Inductive machine learning is a ~~Type of~~ / key task, deriving general principles from data samples.

Components of inductive learning:

Random input vectors(X):

- A generator creates random input vectors.
- These i/p are independently drawn

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from any distribution.

2. System outputs(Y):

- The system produces an output (Y) for each input vector (X).
- This output can be deterministic $y = f(x)$

3. Learning machine:

The learning machines estimates the mapping from input(x) to output(Y') based on observed Samples(X,Y').

General learning Scenario:

Observational setting:

- The learning machine Observes inputs and outputs without controlling them, making it an observed approach.

Inductive learning definition:

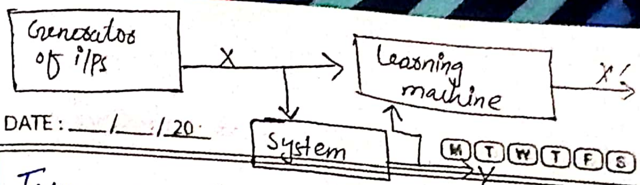
- Estimating unknown- i/p-o/p dependencies from limited observations.
- Data organized into samples.

Learning machine Tasks:

- Form generalization from training data.
- Use a set of functions to approximate system behaviour.
- Finding the best function:

⇒ Given samples $(x_i, f(x_i))$, return a function $h(x)$, that approximates $f(x)$.

⇒ multiple hypothesis $h(x)$ are possible, and the best one is chosen based on criteria like minimum distance from data points.



Types of Approximating functions:

Linear functions:

- Parameters (w) are linear, even if the function appears non-linear (e.g polynomial regression)

Non-linear function:

- Parameters affect the output non-linearily e.g neural network.

Selecting approximating functions:

- Functions $f(X, w)$ are chosen based on prior knowledge about the system.
- Estimating values of parameters (w) is crucial in the learning process.

Statistical dependency vs causality:

- Statistical dependency is about observed relationships, not causality.
- Causality requires external arguments beyond data analysis.

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Loss function:

- measure the difference between actual and predicted output.
- Nonnegative: Large values indicate poor approximation, small values indicate good approximation.

Risk functional:

- Expected value of the loss function over all samples.
- Inductive learning aims to minimize R(w) to find the best approximation.

Importance of human i/p:

- Selection of i/p/o/p variables, data encoding, and incorporating domain knowledge are critical.

Overlap of formal and informal processes:

- Learning involves both formalized algorithms and informal, human oriented decisions.
- Success depends on balancing these aspects.

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SLT

Introduction:

⇒ Structural learning theory, also known as Vapnik-Chervonenkis (VC) theory, is a rigorous framework for finite-sample inductive learning.

⇒ emphasize on mathematical proofs and definitions over practical applications seen in methods like neural networks and Bayesian inference.

⇒ STL focuses on statistical estimation with small samples, balancing model complexity with available data.

Key concepts and principles:

Inductive learning goals

- The goal is to estimate unknown dependencies using available data, aiming to minimize expected risk functional.

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⇒ Since the true risk is unknown, empirical risk is used as an approximation. This principle is called empirical risk minimization (ERM).

Asymptotic Consistency:

⇒ As the number of samples grows, the estimated model should converge to the true model.

⇒ SLT formulates conditions for the consistency of the ERM principle to ensure minimizing empirical risks also minimize true risks.

Growth function and VC Dimension:

⇒ The growth function G(n) describes how the set of approximating functions behaves as the number of samples n increase.

⇒ The VC dimension h indicates where the growth function slows down. VC ensures good generalization.

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Structural risk minimization:

⇒ When the sample size to VC dimension ratio is small, SRM is used instead of ERM.

⇒ SRM involves choosing an optional model complexity for the data set and balancing complexity with data fit to avoid underfitting and overfitting.

Implementation challenges and Optimization approaches:

Stochastic approximations

- Iteratively updates parameters to minimize the risk of function.

Iterative methods:

- Decrease empirical risk iteratively without using gradiant estimates.

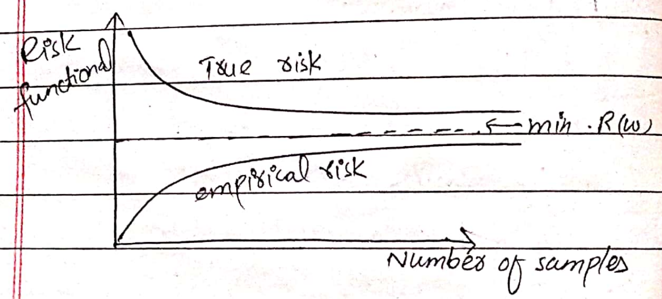
Greedy optimization:

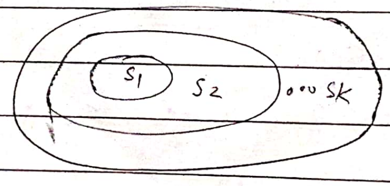
- Optimizes parameter sequentially term by term.

Summary of SLT requirements:

- o Flexible set of approximating functions.
- o A prior knowledge
- o Inductive principle
- o learning method

⇒ SLT provides a comprehensive framework for effective inductive learning, especially useful with finite and small data sets.





Structure of approximating functions

Types of Learning

⇒ Inductive learning methods are divided into two types:

1. Supervised learning → Decision tree, ANN (regression, classification)
2. Unsupervised learning → Genetic algorithm (Clustering, Association)

1. Supervised learning:

⇒ Used to estimate unknown relationships from known input-output samples.

⇒ it involves tasks like classification and regression.

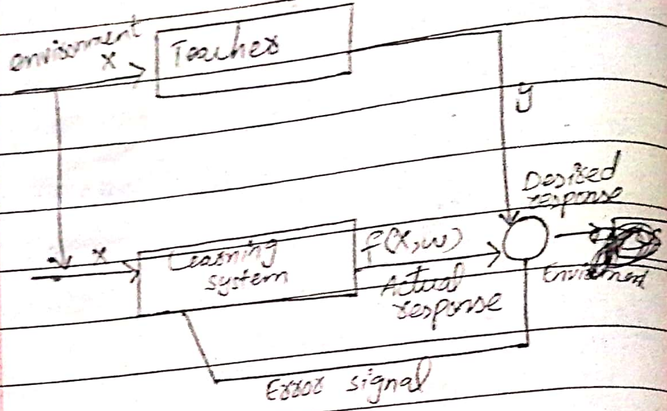
⇒ Supervised means that the output values for training samples are known and provided by a "teacher".

How supervised learning works?

Error surface:

o The performance of the system is measured by mean-square error or the sum of squared errors over training samples.

Block diagram:



Optimization methods:

- Stochastic approximation
- Iterative approach
- Greedy optimization

Applications:

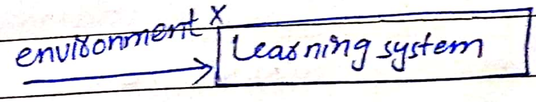
- Pattern classification
- Function approximation
- Logistic-regression
- multi-layered perception
- Decision trees.

2. Unsupervised learning:

⇒ Only input samples are given to the learning system, with no notation of the output during the learning process.

How unsupervised learning works?

Block diagram: ⇒ Self-organized learning



Internal representation:

- The system develops the ability to form internal representation for encoding features of the input examples.
- The representation can be local or global.

Applications:

- Cluster analysis
- Association rule
- ANNs

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Topic:- Common learning Tasks

⇒ The generic inductive learning methods can be subdivided into several common learning tasks:

⇒ These tasks include:

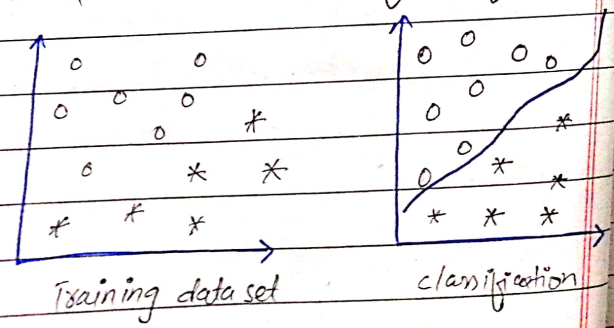
1. Classification:

Def: Grouping or assigns data items into predefined classes.

Complexity:

- In higher dimensions, classification function becomes a hypersurface, which is more complex.

Graphical interpretation of classification:



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2. Regression:

Def: Regression maps data items to a real-value prediction.

- Helps make predictions about numbers.

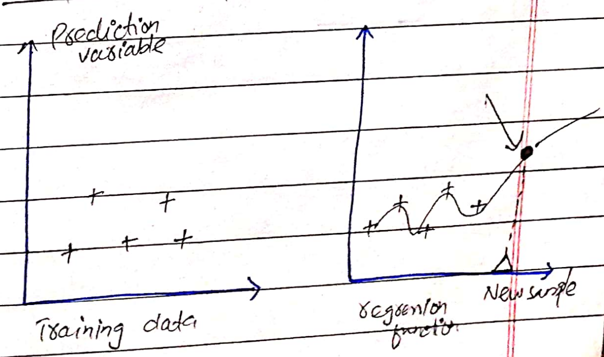
ex:

- If we know how much a student studies and their grades, regression can help predict future grades based on their study time.

Applications:

- To predict continuous outcomes, like prices and Tempratures.

Graphical interpretation:



3. Clustering:

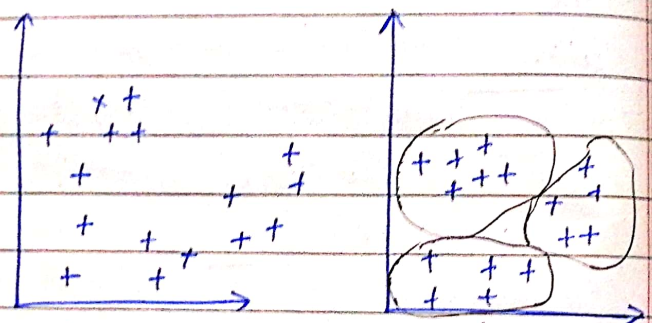
Def:

- Clustering is an unsupervised task that groups data into clusters based on similarity.

Use:

- Clustering helps identify natural grouping within data.

Graphical interpretation:



Training data   Clusters

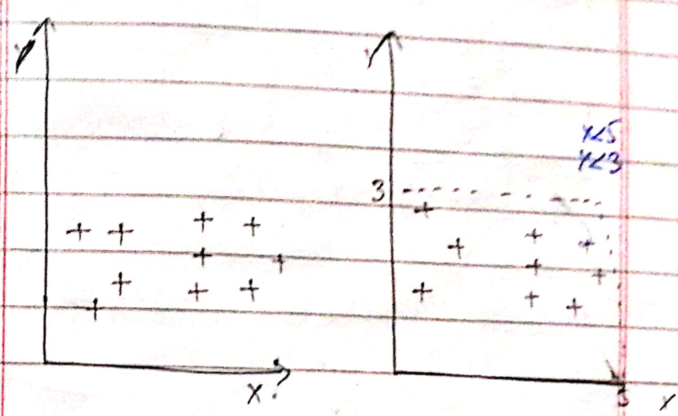
4. Summarization:

Def:

- Provides a compact description of data.
- making a simple, short version of something complex.

Importance:

- It makes large datasets more manageable and understandable.
- Summarization is unsupervised learning.
- method of finding compact description.



5. Dependency modeling:

Def:

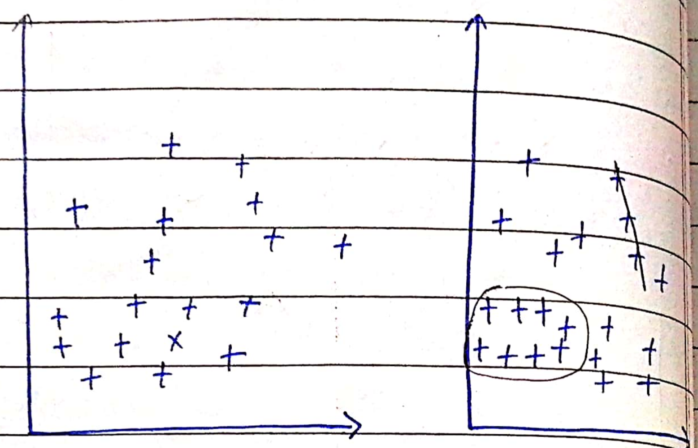
- It figures out how different things are connected in a dataset.

Application:

- Useful in complex systems where global models are too difficult too compute.

- It discovers local models based on a training dataset.

Graphical interpretation:



Training data DLD

6. Change and deviation detection:

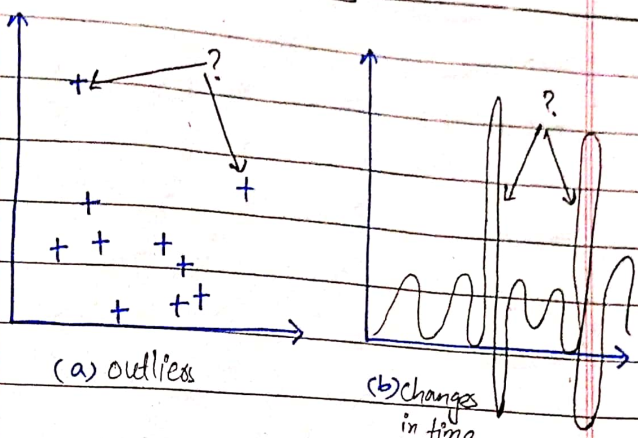
Def:

- Identify significant changes or outliers in data.
- Used to detect outliers.

Use:

- Detecting anomalies
- Changes over time in large datasets.

Graphical interpretation:



(a) outliers (b) Changes in time

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SVMs

Definition:

⇒ Support vector machine was developed by Vladmir vapnik.

⇒ SVM is one of the most popular supervised learning algorithm.

⇒ which is used for classification as well as regression problem.

⇒ Primarily, used for classification in ML. mostly used for binary classification.

Goal:

⇒ The goal of the SVM algorithm is to create the best line or decision boundary that can segregate n-dimensional space into classes so that easily put the new data point in the correct category in the future.

mean the best hyperplane with largest distance b/w 2 classes in N-Dimensional space

⇒ can be used for:

- Face detection
- image classification
- Text categorization

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Key Concepts

Support vectors:

⇒ Data points that are closest to the hyperplane is called support vectors.

⇒ Separating line will be defined with the help of these data points.

Hyperplane: — best boundary from all decision boundary

that separate different classes or groups of data points.

⇒ it is a decision plane or space / boundary which is divided between a set of objects having different classes.

in higher dimensional space

⇒ $wx+b=0$

Margin:

⇒ Distance between support vector or -ive and +ive hyperplane and hyperplane

⇒ Gap between two lines on the closet data points of different classes.

⇒ Calculated as perpendicular distance from the line to the SV.

⇒ Large margin considered as good margin and small margin considered as a bad margin.

Hard margin:

⇒ The maximum-margin hyperplane or the hard margin hyperplane is a hyperplane that properly separates the data points of different categories without any misclassifications.

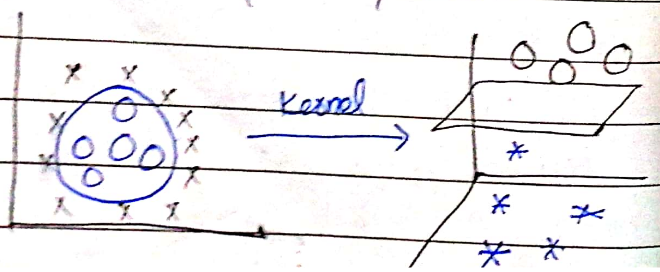
Soft margin:

⇒ When the data is not perfectly separable or contains outliers, SVM permits a soft margin techniques.

Kernel function:

⇒ Kernel function takes low dimensional i/p space and transform it into a higher dimensional space.

⇒ It converts not separable problem to separable problem.



⇒ Used when the data points are not linearly separable.

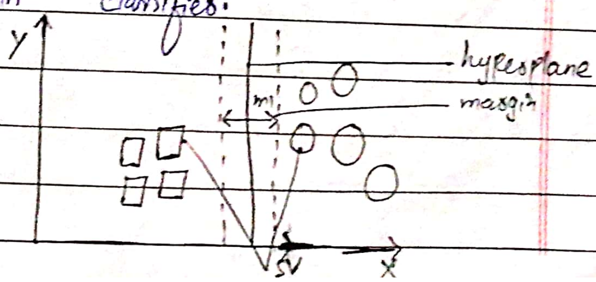
⇒ Some of the common kernel functions:

- Polynomial
- radial-bias function
- Sigmoid

Types of SVM

Linear SVM:

⇒ Used for linearly separable data, which means if a dataset can be classified into two classes by using a single straight line, then such data is termed as linearly separable data, and classifier is used called as linear svm classifier.



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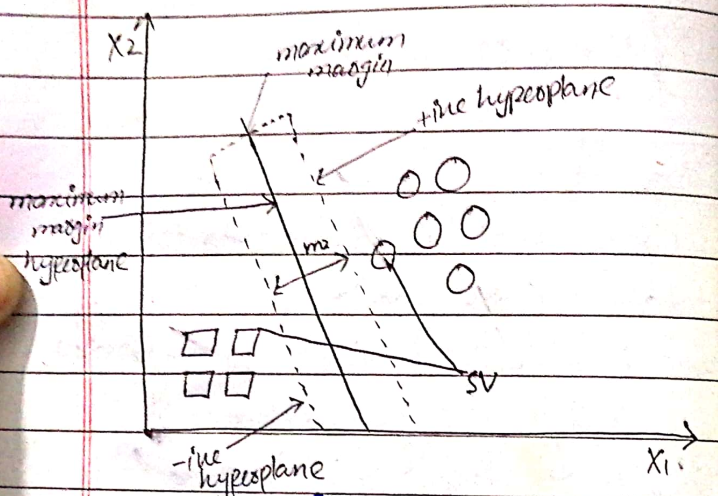
Non Linear SVM:

⇒ Used for non-linearly separate data meas if a dataset cannot be classified by using a straight line then such data is termed as non-linear data and classifier is called non-linear svm classifier.



SVM algorithm steps

⇒ The basic steps of svm are:

- Select Two hyperplanes.
- maximize their distance (margin)
- The average line will be the decision boundary.

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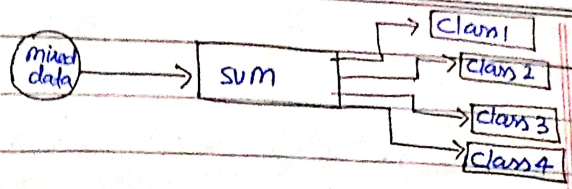


Pros:

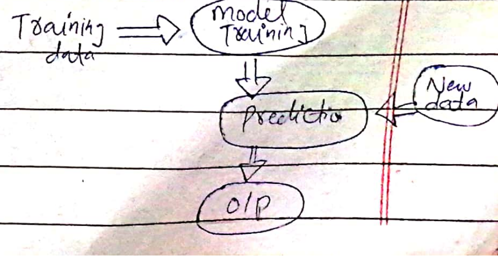
- Highly accurate
- handle many features
- Speed is fast
- requires less time.
- Robustness to noise
- versatality
- sparsity
- control of complexity

Cons:

- scalability with larger datasets
- Speed and size requirement both in training and machine algorithm is more.
- lack of probabilistic interpretation.

Applications:

- face detection
- text & hypertext arrangement
- Handwriting remembrance
- speech recognition.



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KNN

⇒ Stands for nearest neighbour algorithm.

⇒ Simplest algorithm based on supervised machine learning.

⇒ KNN algorithm assumes the similarity between the new case /data and available cases and put the new case into the category that is most similar to the available categories.

⇒ KNN can be used for regression as well as classification but mostly it is used for the classification problems.

⇒ Non-parametric, which means it does not make any assumptions on underlying data.

Local decision boundary:

⇒ Unlike SVM, which use a global model, KNN determines the decision boundary locally

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based on nearby data points.

Lazy learning:

- Also called lazy learner because it does not learn from the training set immediately instead it stores the dataset and at time of classification, it performs an action on dataset.

- in training Phase just stores data.

Selecting K value:

- There is no particular way to select K.
- A very low value of K such as K=1 or K=2, can be noisy and lead to the effects of outliers in the model.
- Large values for K is good.
- Often chose based on experience or testing. it's better if K is odd to avoid ties.
- Common choices are 3 or 5.

Time complexity:

- The time complexity is linear with the size of the training set.

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as it requires distance calulations between the test sample and all the training samples.

Distance measure:

⟹ The choice of distance measure is crucial, especially as the number of features increases.

- Euclidean
- Cosine. etc

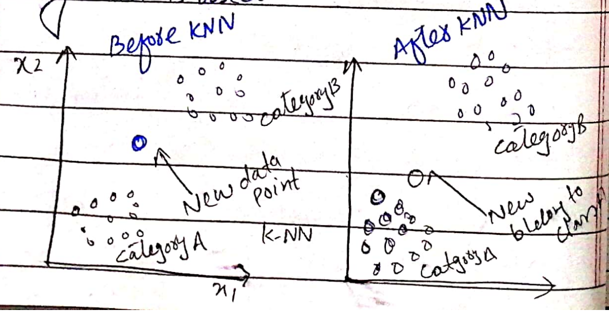
Why we need K algorithm?

⟹ Suppose there are two categories

- Category A
- category B

⟹ We have a new data point x_1. So this data point lie in which of these categories.

To solve this problem, KNN algorithm is used.



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How does KNN works?

⟹ There are the following steps:

Step-1:

- Select the value of K of the neighbours.

Step-2

- Calculate the Euclidean distance of K number of neighbours.

Step-3

- Take the K nearest neighbours as per the calculated Euclidean distance.

Step-4

- Among these k neighbours, Count the number of data points in each category.

Step5

- Assign the new data points to that category for which the number of neighbour is maximum.

Step-6:

- Now, model is ready.

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Advantages:

- Simple to implement
- Robust to noisy data.
- can be more effective if the training data is very large.

Disadvantages:

- Always needs to determine the value of k which may be complex sometime.
- Computation cost is high

Applications:

⇒ Pattern recognition
⇒ Recommendation system
⇒ medical diagnosis
⇒ Finance
⇒ Anomaly detection
⇒ Geospatial analysis
⇒ Text classification

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Example of KNN:

Query ⇒ $x = (maths = 6, CS = 8)$,
$k = 3$

maths	CS	Result
4	3	fail
6	7	pass
7	8	pass
5	5	fail
8	8	pass

$$= \sqrt{|X_{01} - X_{A1}|^2 + |X_{02} - X_{A2}|^2}$$

$$I = \sqrt{(6-4)^2 + (8-3)^2} = 5.38$$

$$II = \sqrt{(6-6)^2 + (8-7)^2} = 1$$

$$III = \sqrt{(6-7)^2 + (8-8)^2} = 1$$

$$IV = \sqrt{(6-5)^2 + (8-5)^2} = 3.15$$

$$V = \sqrt{(6-8)^2 + (8-8)^2} = 2$$

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⇒ Probability of pass = 3

⇒ probability of fail = 0

So, Given sample will result Column is pass.

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Model selection VS Generalization:

model selection:

⇒ The process of choosing the best model among a set of candidates model based on their performance on a given dataset.

Generalization:

⇒ The models ability to perform well on unseen data that was not part of the training data.

Validation:

⇒ Ensuring that the model meets the user-defined objectives and accurately represents the system.

Verification:

⇒ Confirming the model is correctly built from the data with sufficient accuracy.

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Methods for validation in inductive learning tasks:

1. Resubstitution:

⇒ use all data for both training and testing.

Pros:

- Simple

Cons:

- Optimistically based

2. Holdout method:

⇒ Dataset is divided into two parts:

- one for training
- one for testing

Pros:

- simple
- independent sets

Cons:

- Different split yields different estimate
- Not suitable for small datasets.

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3. K-fold cross validation:

⇒ The dataset is divided into k equally sized folds.

⇒ The model is trained k times, each time using k-1 folds for training and the remaining folds for testing.

Pros:

- Reliable
- utilize the entire datasets.

Cons:

- Computationally expensive

4. Leave-one-out-cross validation:

⇒ A special case of k-fold cross-validation where k is equal number of data points in the dataset.

⇒ Each data point is used once as a test and the remaining used for training.

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Pros:

- Use all data
- No data is wasted.

Cons:

- very high computational cost.

5. Bootstrap method:

⟹ Resample data with replacement to create multiple datasets.

Pros:

- Useful for small datasets.
- Better error estimates

Cons:

- High computational effort.
- Depends on resampling strategy

6. Repeated random sampling:

multiple random splits into training/testing sets.

Pros:

- Robust estimate
- flexible

Cons:

- High variance
- requires many repetitions.